Analysis 1: Aug-Nov 2026

Definition: $\langle a_k : k \geq 1\rangle$ be a given sequence. We say that the series $\sum_k a_k$ converges to θ if the sequence of partial sums s_n defined via

$$s_n = \sum_{k=1}^{n} a_k, \quad 1 \leq n < \infty$$

satisfies $\forall \epsilon > 0$, $\exists n^*$ such that $n \geq n^*$ implies $|s_n - \theta| < \epsilon$, (*i.e.* converges to θ). The series is said to be convergent if it converges to some $\theta \in \mathbb{R}$.

You may use that for $0 < r < 1$, the series $\sum_n r^n$ is convergent. You may also use that a sequence is convergent if and only it is a Cauchy Sequence. Any other fact that you want to use, you must state precisely and prove the same.

Exercises set #3

Let $\langle a_k : k \geq 1\rangle$, $\langle b_k : k \geq 1\rangle$, $\langle c_k : k \geq 1\rangle$ be sequences such that the series $\sum_k a_k$, $\sum_k b_k$ and $\sum_k c_k$ are convergent.

(1) Show that the sequence $\langle c_k : k \geq 1\rangle$ converges to 0.

(2) Give an example of a sequence $\langle d_k : k \geq 1\rangle$ that converges to 0 but the series $\sum_k d_k$ does not converge.

(3) Let $\langle q_k : k \geq 1\rangle$ defined by $q_k = a_k^3$. Show that $\sum_k q_k$ is a convergent series. Let $t_k = a_k^4$, $k \geq 1$. Explore if we can conclude that $\sum_k t_k$ is a convergent series?

(4) Let $\langle u_k : k \geq 1\rangle$ be defined by $u_k = 133a_k, \ \ k \geq 10^9$ and $u_k = k^3$ for $k < 10^9$. Show that $\sum_k u_k$ is a convergent series.

(5) Let $\langle v_k : k \geq 1\rangle$ be defined by

$$v_k = 823b_k + a_k, \quad k \geq 1.$$

Show that $\sum_k v_k$ is a convergent series.

(6) Suppose $a_k \geq 0, \ b_k \geq 0$ for all $k \geq 1$. Let $\langle w_k : k \geq 1\rangle$ be defined by

$$w_k = 703c_k - 88a_k * b_k, \quad k \geq 1.$$

Show that $\sum_k w_k$ is a convergent series. Given an example to show that the conclusion may not be true if we drop the assumption that $a_k \geq 0, \ b_k \geq 0$ for all $k \geq 1$.

(7) For $k \geq 1$ and $2^{k-1} \leq n < 2^k$, let $x_n = 2^{-2k}$. Show that $\sum_n x_n$ is a convergent series.

(8) Show that if $f_n = \frac{1}{n^2}$, then $\sum_n f_n$ is convergent.

(9) Let $\alpha > 1$ and for $k \geq 1$, $2^{k-1} \leq n < 2^k$, let $y_n = 2^{-\alpha k}$. Show that $\sum_n y_n$ is a convergent series.

(10) Show that if $h_n = \frac{1}{n^\alpha}$, for some fixed $\alpha > 1$, then $\sum_n h_n$ is a convergent series.

(11) Let $\langle z_k : k \geq 1\rangle$ be a sequence such that $z_k > 0$ for all k and $\limsup_k \frac{z_k}{z_{k+1}} = r < 1$. Show that $\sum_n z_n$ is a convergent series.

(12) Show that if $r > 1$ in the previous question, $\sum_n z_n$ does not converge, while if $r = 1$ then the test is inconclusive, namely, construct two examples where $r = 1$, but first is a convergent series and the second is a divergent series.

(13) Show that for any x, $z_k = \frac{x^{k-1}}{n!}$, $k \geq 1$, $\sum_k z_k$ is a convergent series.